

Heat treatment of steels

Hardening

Direct hardening

Austenitize and Quench

Selective austenitize and Quench

- Flame
- Induction
- Laser
- Electron Beam

Diffusion treatment

Carburize(Diffuse C)

Nitride(Diffuse N)

Boronize(Diffuse B)

Carbonitriding (Diffuse C +N)

Ferritic Nitro carburizing (Diffuse C +N)

Softening

Recrystallization

Annealing

- Process
- Full
- Spherodizing
- Normalizing

Tempering

- Austempering
- Martempering

Heating for Hot working

Conditioning

Stress relieving

Spring aging

Steam treating

Cryogenic treatment

- **Hardening:**
- Hardening of steels is done to increase the strength and wear properties.
- **Softening:**
 - Softening is done to reduce strength or hardness, remove residual stresses, improve toughness, restore ductility, refine grain size or change the electromagnetic properties of the steel.
- Restoring ductility or removing residual stresses is a necessary operation when a large amount of cold working is to be performed, such as in a cold-rolling operation or wire drawing.
- **Material Modification:**
- Heat treatment is used to modify properties of materials in addition to hardening and softening. These processes modify the behaviour of the steels in a beneficial manner to maximize service life.

Quenching

- Quenching is the act of rapidly cooling the hot material in water, oil or air to harden the material to obtain certain material properties.
- Quenching prevents undesired low-temperature processes
- Quenching can reduce the crystal grain size of both metallic and plastic materials, increasing their hardness. In metallurgy, quenching is most commonly used to harden steel.
- Water:
 - Good rapid quenching medium,
 - Corrosive with steel,
 - Cause distortion or cracking.
- Salt Water:
 - More corrosive than plain water
 - Must be rinsed off immediately.



Quenching

❖ Oil:

- Slower cooling rate
- Reduces the likelihood of cracking.
- Results in fumes, spills, and sometimes a fire hazard.



❖ Polymer quench:

- Cooling rate in between water and oil, it can vary by alter the mixture of water and glycol polymers
- Repeatable results with less corrosion and fire hazard

❖ Cryogenic Quench:

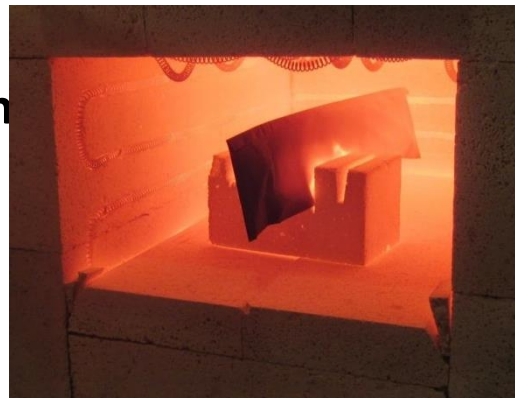
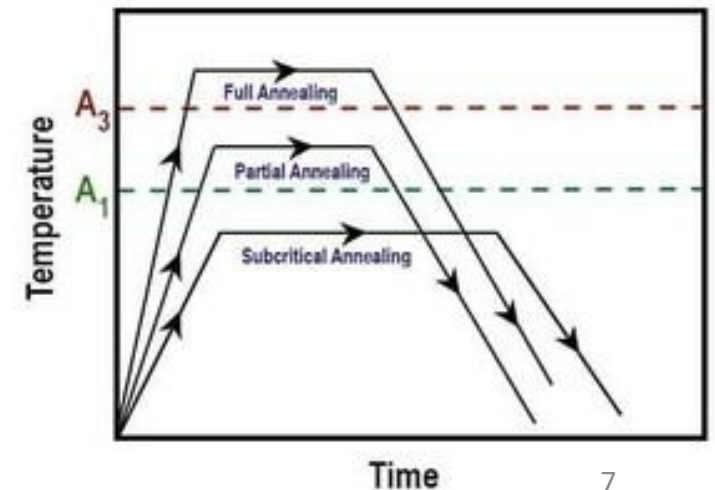
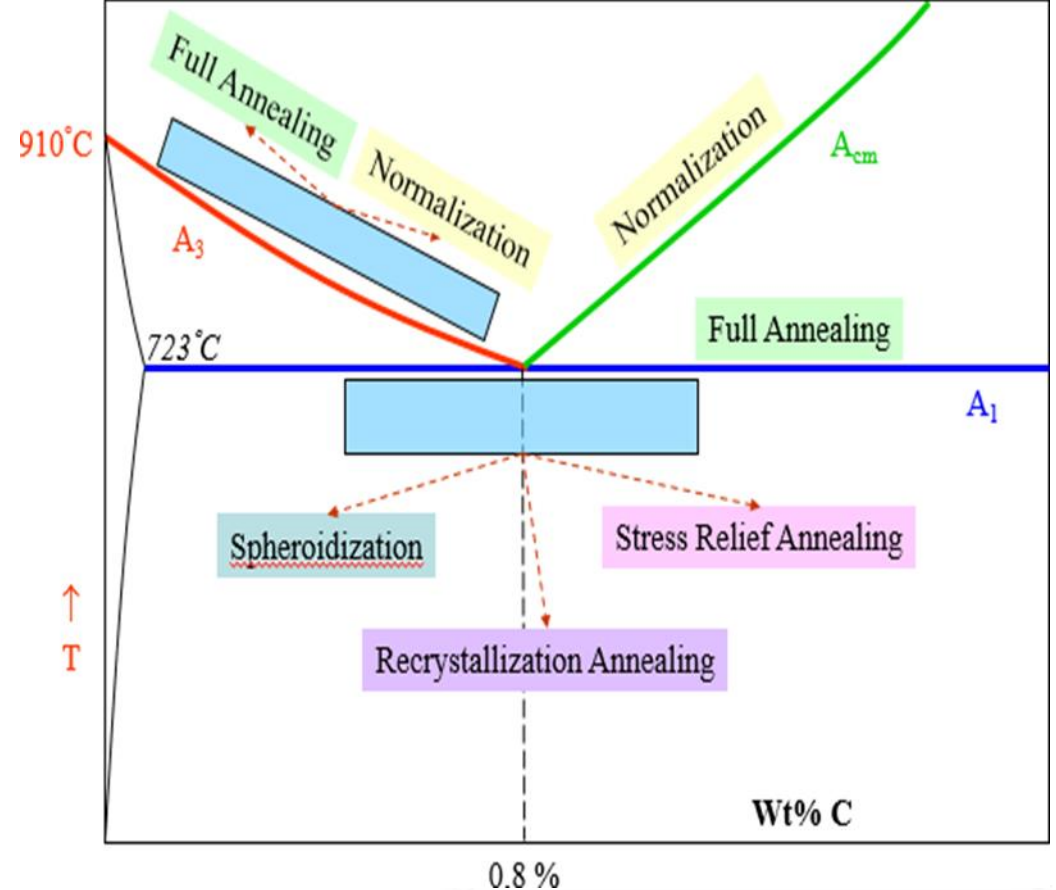
- Function of the lowest temperature
- Fully martensite to avoid cracking and dimensional instability
- To eliminate retained austenite in high carbon and high alloy steel

Various Heat Treatment Processes

- **Normalizing**
- **Annealing.**
- **Hardening.**
- **Tempering**
- **Case hardening**
 - ✓ **Carburizing**
 - ✓ **Cyaniding**
 - ✓ **Nitriding**
- **Surface hardening**
 - ✓ **Induction hardening,**
 - ✓ **Flame hardening.**

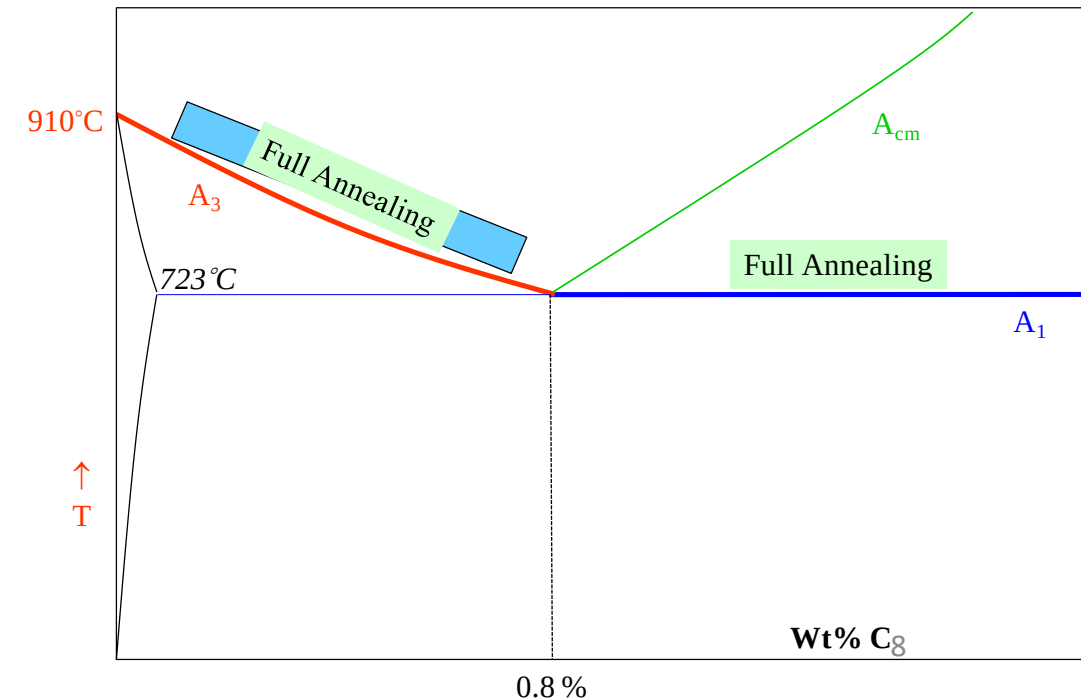
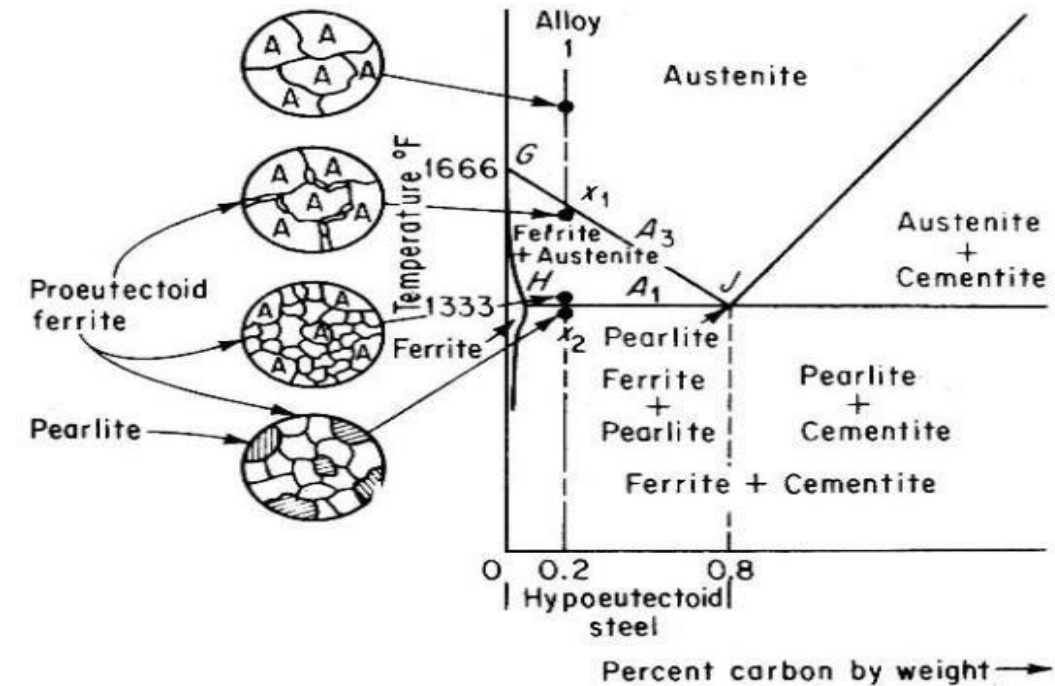
Annealing

- Material is exposed to an elevated temperature but below critical temperature for an extended time period and then slowly cooled at still air, allowing phase changes.
- Purpose
 - Alteration of ductility and toughness
 - Induction of softness
 - Refinement of grain size and
 - Removal of gases and stresses
- Disadvantage: Material losses its strength and hardness
- Utilized for low- and medium-carbon steels.
 - Full Annealing
 - Process Annealing or Stress Relief Annealing
 - Spheroidising
 - Normalizing



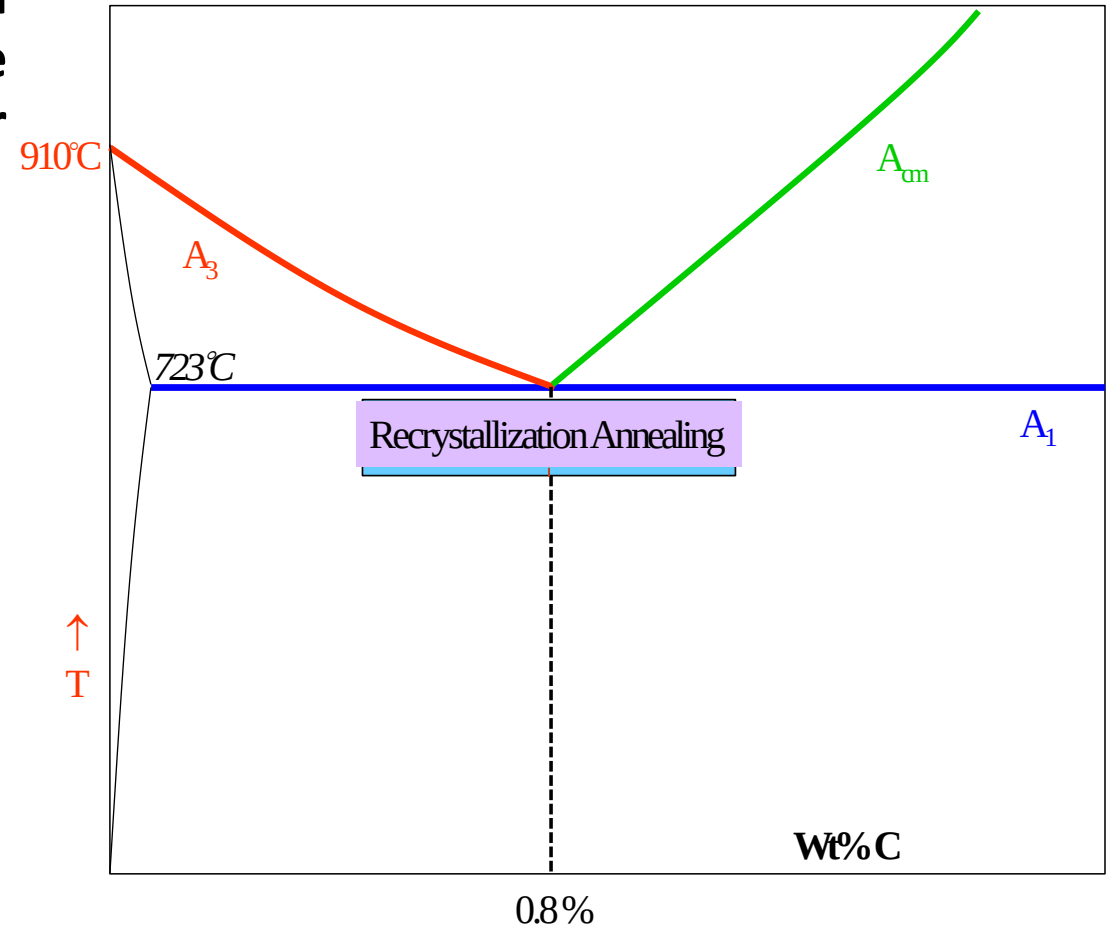
Full annealing

- Full annealing is the process of slowly raising the temperature about 50°C above the Austenitic temperature line A_3 , or above A_1 line into the Austenite-Cementite region.
- It is then slowly cooled at the rate of about 20°C/hr in a furnace to about 50°C into the Ferrite-Cementite range.
- The grain structure has coarse Pearlite with ferrite or Cementite (depending on whether hypo or hyper eutectoid).
- The steel becomes soft and ductile.



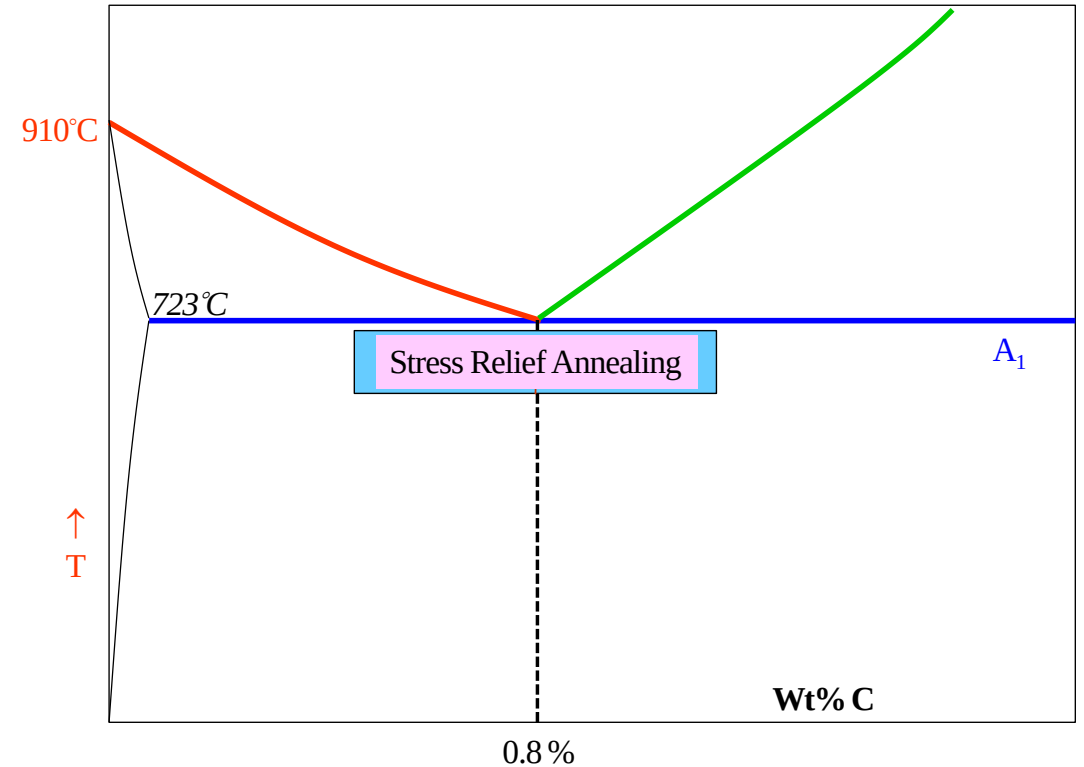
Process Annealing

- Process Annealing is used to treat work-hardened parts made out of low-Carbon steels (< 0.25% Carbon). This allows the parts to be soft enough to undergo further cold working without fracture
- Process annealing is done by raising the temperature to just below the Ferrite-Austenite region, line A_1 on the diagram.
- This is held at that temperature, long enough to allow recrystallization of the ferrite phase, and then cooled in still air.
- The only change that occurs is the size, shape and distribution of the grain structure.



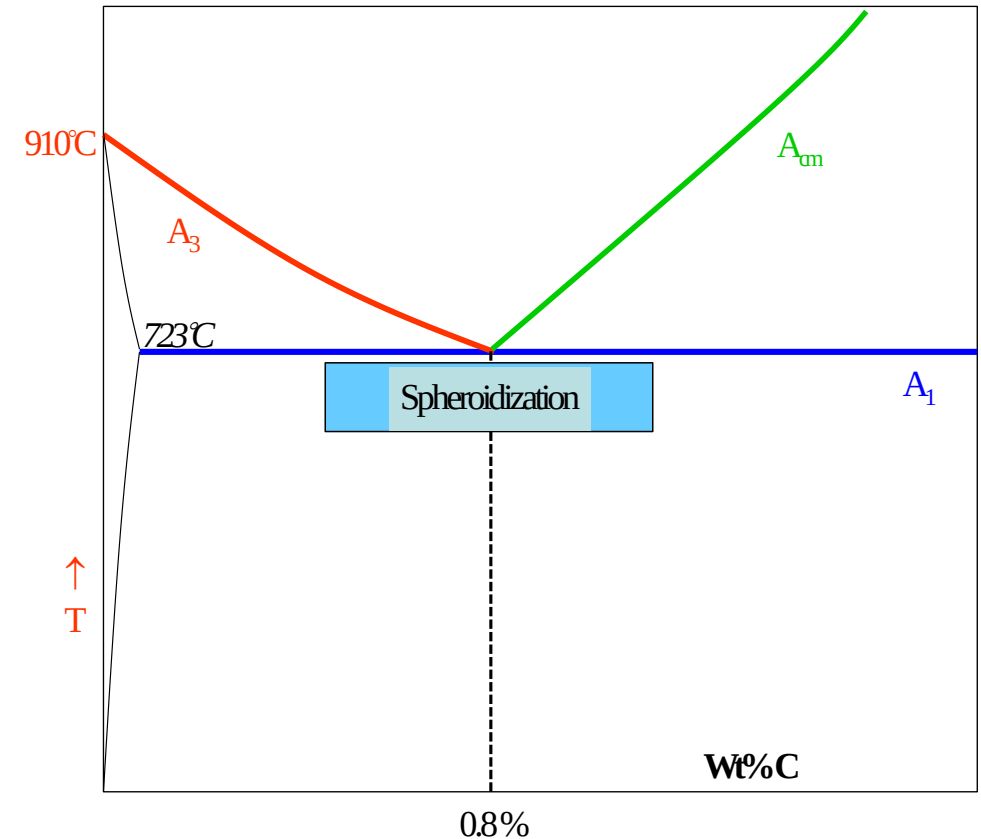
Stress relief anneal

- Stress relief anneal is used to reduce residual stresses in large castings, welded parts and cold- formed parts.
- Such parts tend to have stresses due to thermal cycling or work hardening.
- Parts are heated to temperatures of up to 600 - 650 °C (1112 - 1202 °F), and held for an extended time (about 1 hour or more) and then slowly cooled in still air.



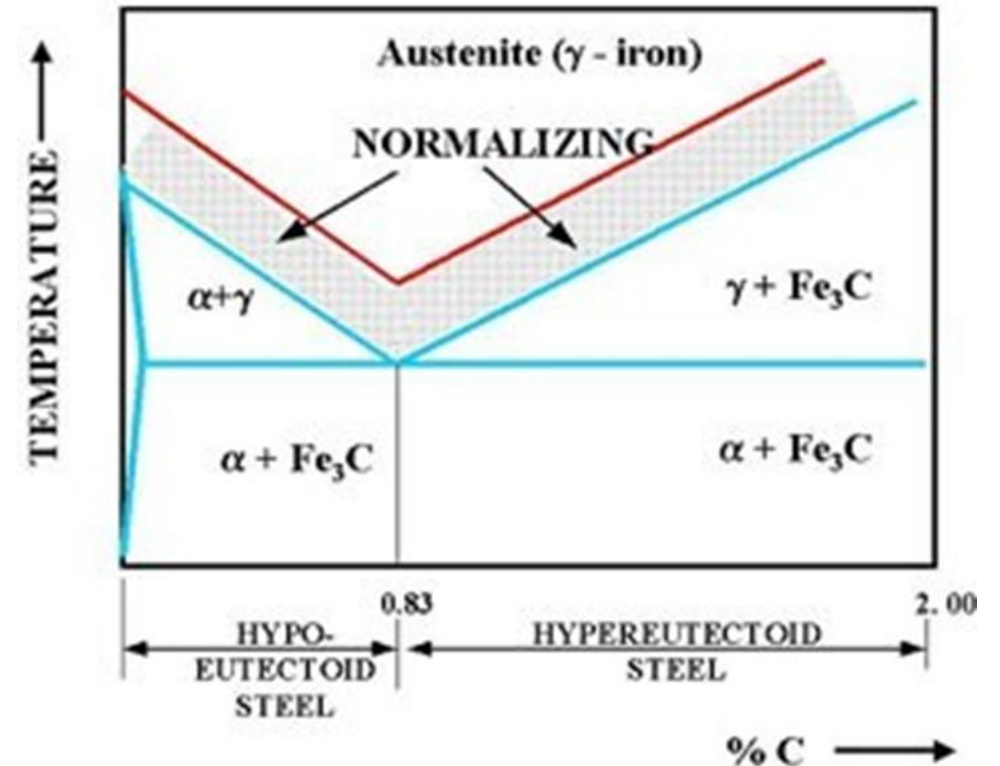
Spheroidization

- Spheroidization is an annealing process used for high carbon steels (Carbon > 0.6%) that will be machined or cold formed subsequently.
- This is done by one of the following ways:
 - Heat the part to a temperature just below the Ferrite-Austenite line, line A_1 or below the Austenite-Cementite line.
 - Hold at that temperature for a prolonged time followed by fairly slow cooling.
- The main purpose of the treatment is to increase the ductility of the sample.
- Spheroidization also improves resistance to abrasion.



Normalizing

- Normalizing is the process of raising the temperature to over 60 (140 °F), above line A_3 or line A_{CM} fully into the Austenite range.
- It is held at this temperature to fully convert the structure into Austenite, and then removed from the furnace and cooled at room temperature under natural air convection.
- This results in a grain structure of fine Pearlite with excess of Ferrite or Cementite. The resulting material is soft.
- This process is considerably cheaper than full annealing since there is no added cost of controlled furnace cooling.
- The main difference between full annealing and normalizing is that fully annealed parts are uniform in softness because of the controlled furnace cooling.
- In the case of the normalized part, the cooling is non-uniform resulting in non-uniform material properties across the part.
- This may not be desirable if further machining is desired, since it makes the machining job somewhat unpredictable.
- In such a case it is better to do full annealing.



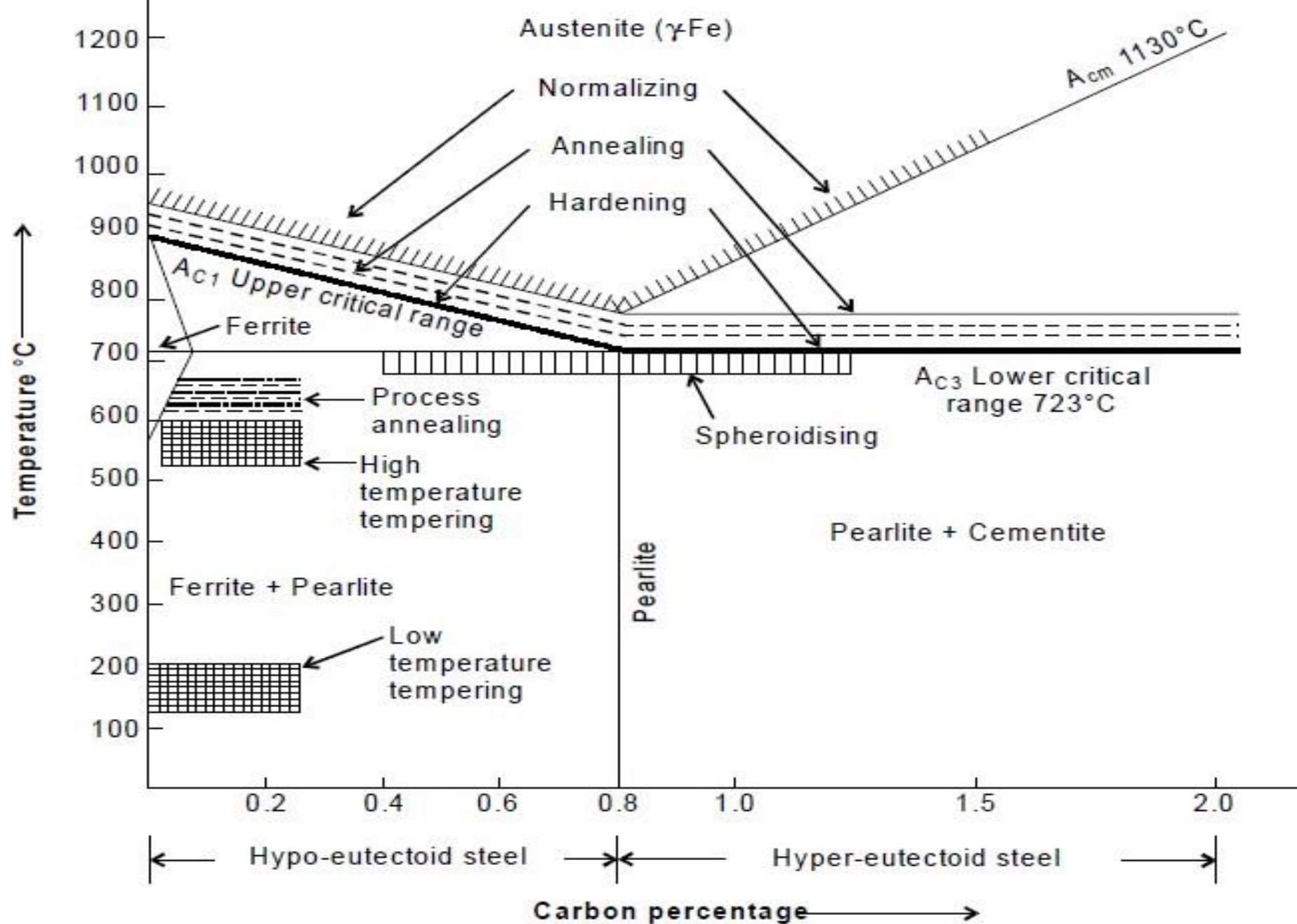


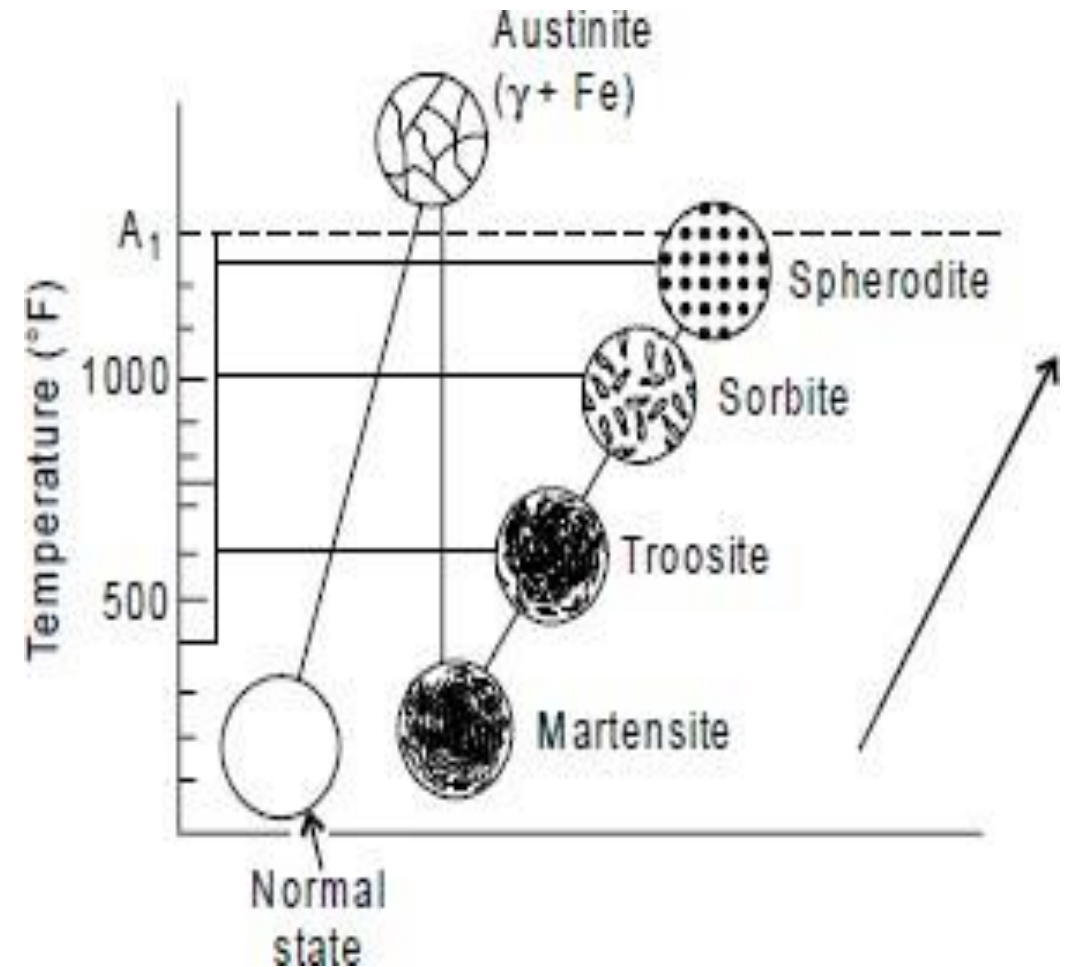
Fig. 8.7 Heating temperature ranges for various heat treatment processes

Tempering

- **Steel is usually harder than necessary and too brittle for practical use after being hardened.**
- **Severe internal stresses are set up during the rapid cooling of the metal.**
- **Steel is tempered after being hardened to relieve the internal stresses and reduce its brittleness.**
- **Tempering consists of heating the metal to a specified temperature and then permitting the metal to cool.**
- **The rate of cooling usually has no effect on the metal structure during tempering. Therefore, the metal is usually permitted to cool in still air.**
- **Temperatures used for tempering are normally much lower than the hardening temperatures.**
- **The higher the tempering temperature used, the softer the metal becomes.**
- **High-speed steel is one of the few metals that becomes harder instead of softer after it is tempered.**

Tempering

- In tempering is done on steel after hardening
- Steel is reheated to a temperature below the lower critical temperature and then followed by a desired rate of cooling.
- After tempering structure is changed into secondary structure like troostite, sorbite and spheroidite.

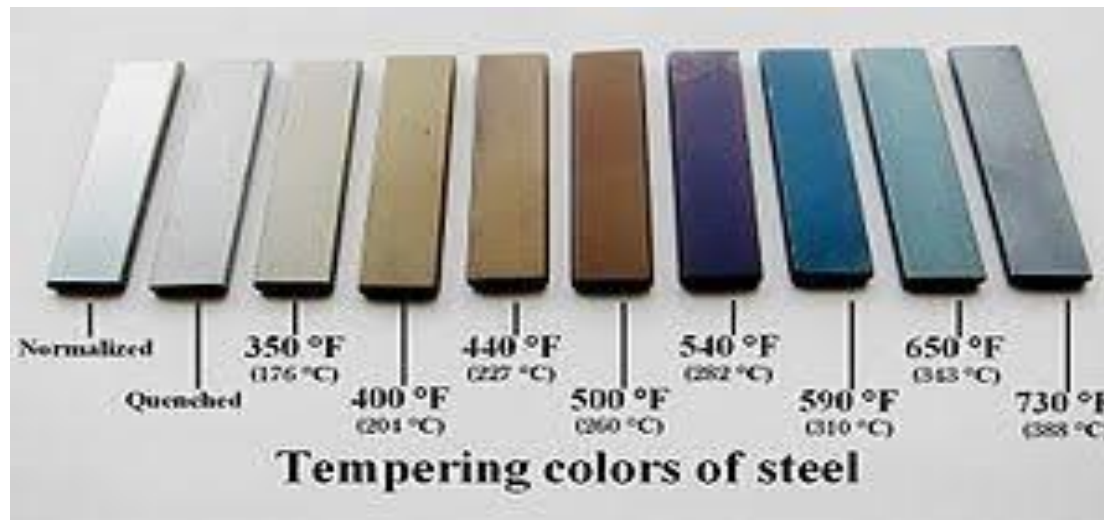


Tempering

- The mechanism of tempering depends on the steel and the tempering temperature. The prevalent Martensite is a somewhat unstable structure.

When heated, the Carbon atoms diffuse from Martensite to form a carbide precipitate and the concurrent formation of Ferrite and Cementite, which is the stable form.

- Tool steels for example, lose about 2 to 4 points of hardness on the Rockwell C scale.
- Toughness (as measured by impact strength) is increased substantially. Springs and such parts need to be much tougher, these are tempered to a much lower hardness.



Austempering is an isothermal process to achieve a solely bainitic structure.

This is accomplished by heating the part within the austenite range and then quenching the part into a bath of hot oil or molten salt held at a constant temperature of 260-400°C or 500-750°F (above the Ms temperature of the alloy).

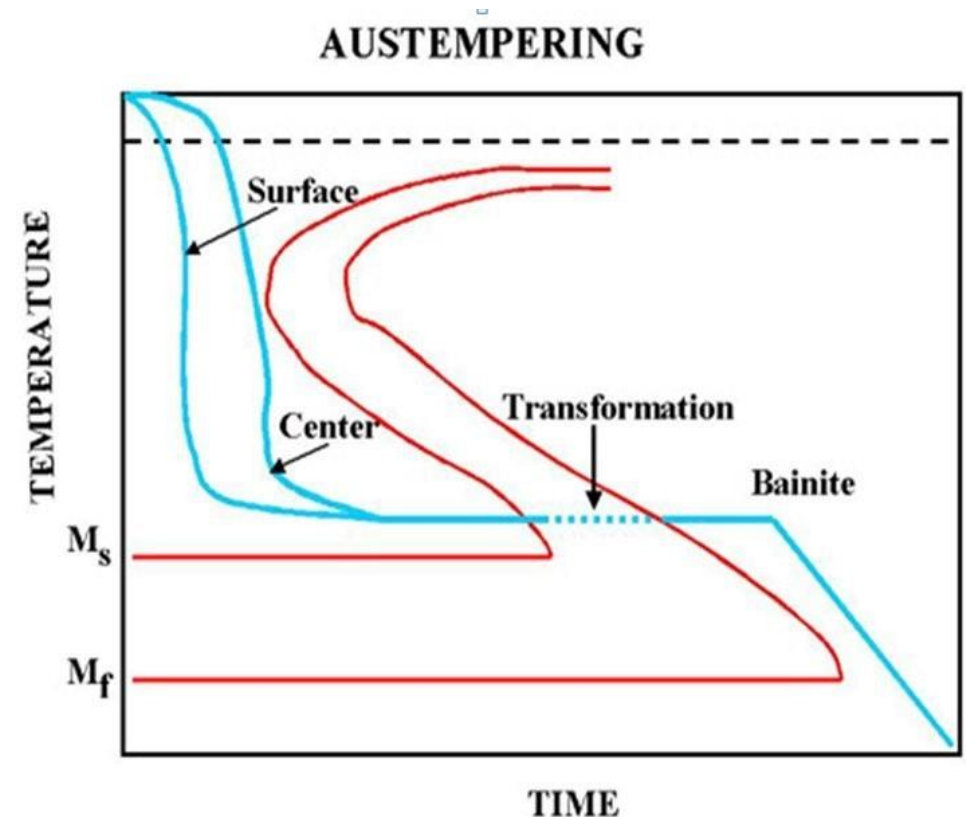
The part is then allowed to transform isothermally to achieve a bainitic structure, and allowed to cool in a convenient manner, usually in air.

As the part is held longer at this temperature, the Austenite transforms into fine bainite structure in steel but with minimum distortion and residual stresses.

Bainite is tough enough so that further tempering is not necessary, and the tendency to crack is severely reduced.

Austempering

- Austempering is a quenching technique that is applied to ferrous metals, most notably steel and ductile iron.
- The part is not quenched through the Martensite transformation.
- Instead the material is quenched above the temperature when Martensite forms (M_s), around 315°C.
- It is held till at this temperature till the entire part reaches this temperature.
- As the part is held longer at this temperature, the Austenite transforms into fine bainite structure in steel but with minimum distortion and residual stresses.
- Bainite is tough enough so that further tempering is not necessary, and the tendency to crack is severely reduced.



Advantages Of Aus-tempering

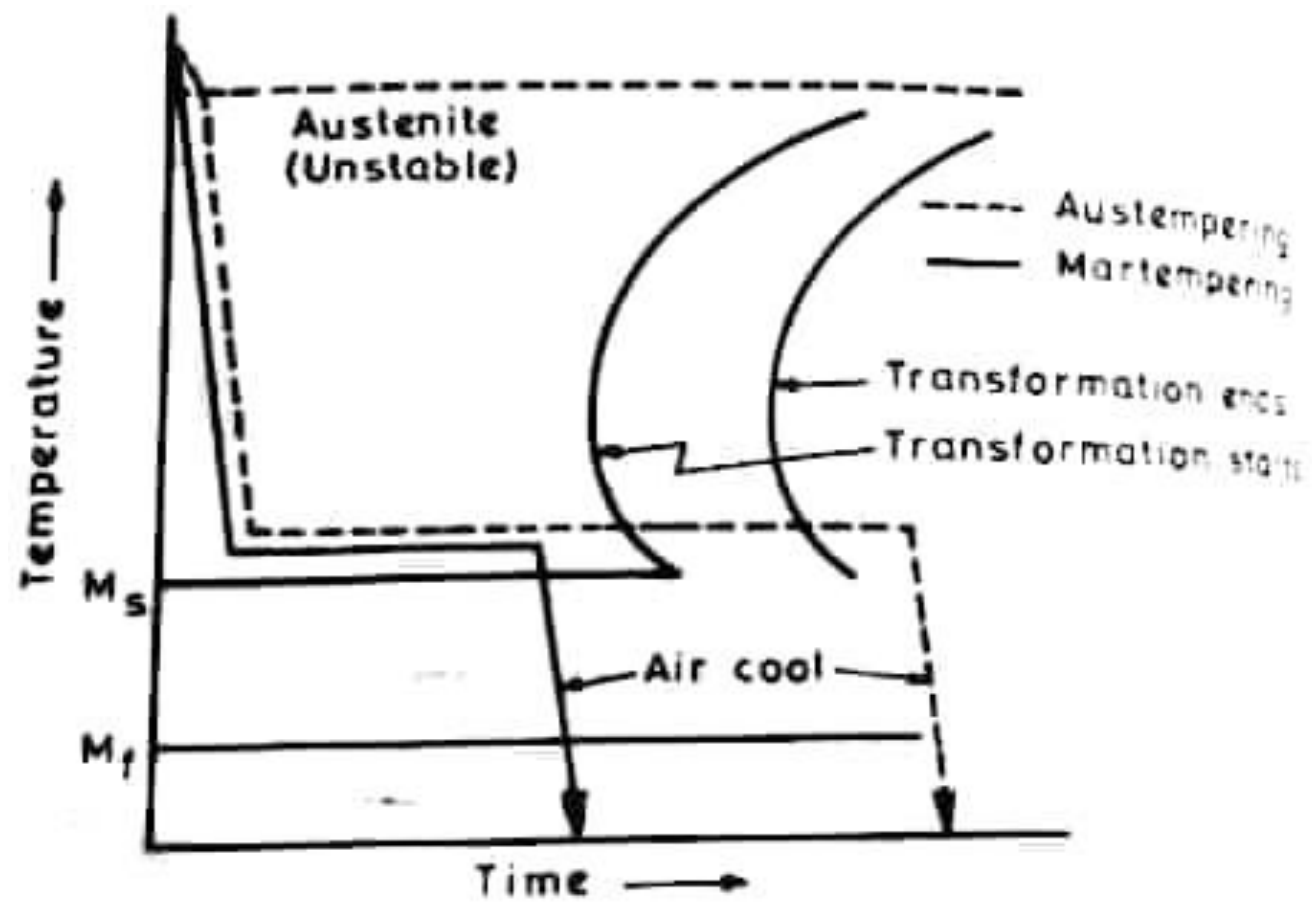
- ✓ Quenching cracks are avoided.
- ✓ Distortion and warping are avoided.
- ✓ A more uniform microstructure is obtained.
- ✓ Mechanical properties of bainite are superior to conventional hardening microstructure.

Limitations of Aus-Tempering

- ✓ The process is very costly.
- ✓ The process is time consuming

Martempering

- Martempering is a heat treatment for steel involving austenitisation followed by step quenching, at a rate fast enough to avoid the formation of ferrite, pearlite or bainite to a temperature slightly above the martensite start (M_s) point.
- The drawback of this process is that the large section cannot be heat treated by this process



Austempering & Martempering

After heating, pieces are quenched in salt bath where the temperature of the bath is set just above the given material's martensitic start point.

Compared to austempering, where quenches run at higher temperatures, martempered parts are quenched at a cooler temperature (generally around 350°F, depending on the alloy).

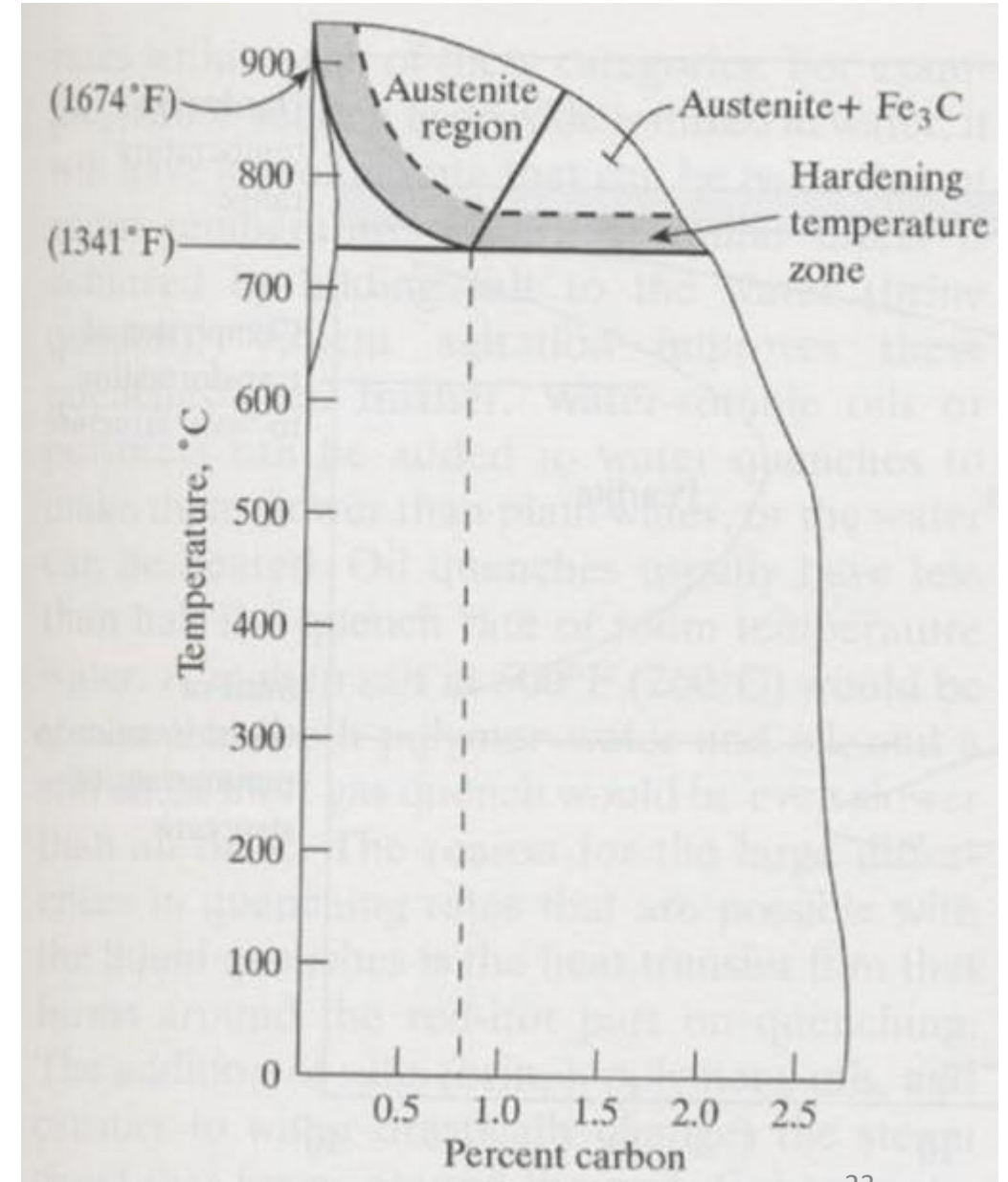
The parts are held in the quench at this temperature to cool only long enough—between five and ten minutes—for the core and surface temperatures to equalize. The parts are then pulled from the quench and further cooled in still air at room temperature.

Quenching the part only long enough to stabilize surface and core temperatures and then cooling in still air promotes the formation of martensite, the hardest form a steel.

Forming martensite using other heat treatment processes is possible, but parts encounter significantly more stress in those processes, which heightens the risk of distortion.

Direct Hardening

- Hardness is a function of the carbon content of the steel. (0.6% or above C)
- Hardening of a steel requires a change in structure from the BCC to FCC structure found in the austenitic region.
- When suddenly quenched, the martensite is formed. This is a very strong and brittle structure.
- When slowly quenched it would form austenite and pearlite which is a partly hard and partly soft structure.
- When the cooling rate is extremely slow then it would be mostly pearlite which is extremely soft.



Effects Of Heat Treatment

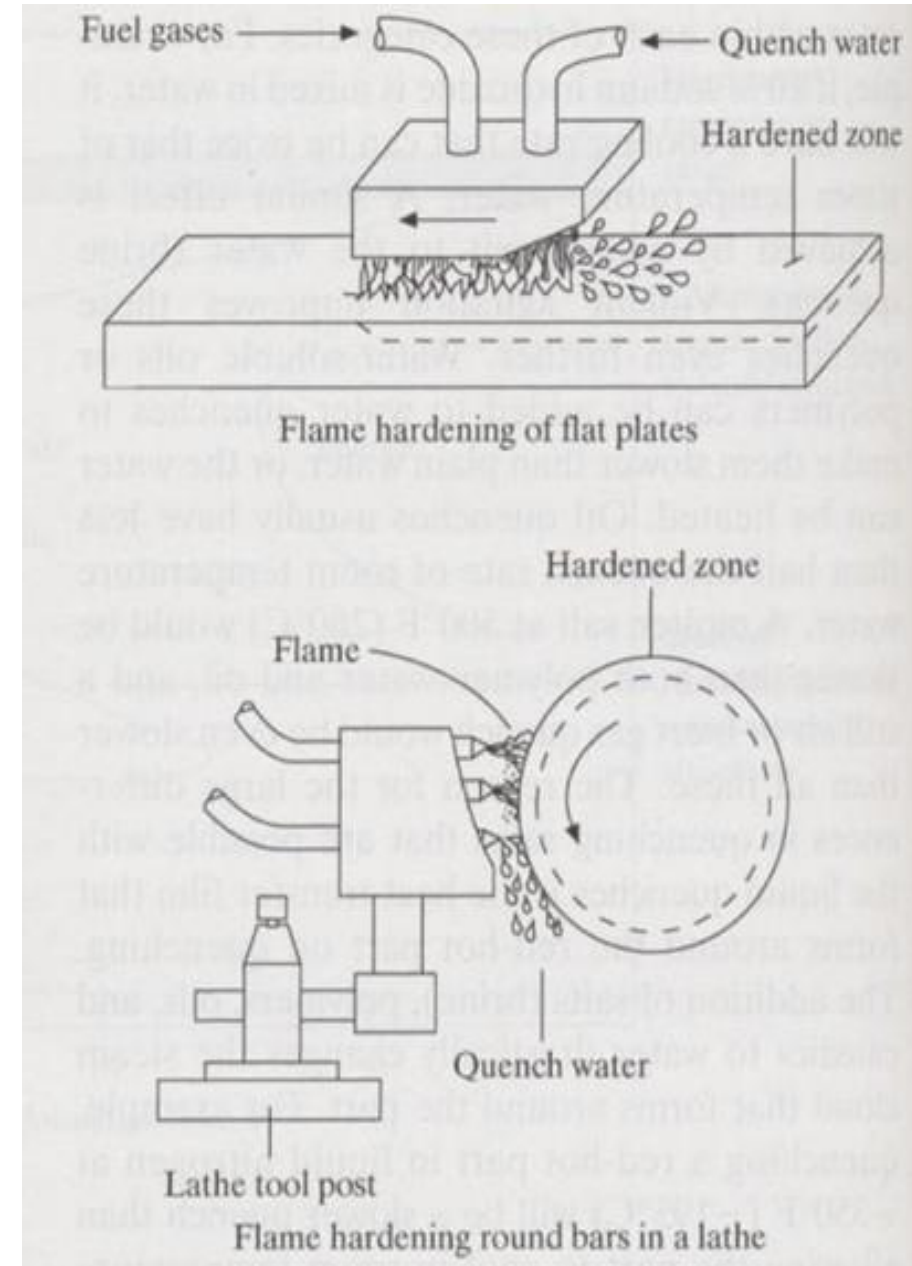
Annealing & Normalizing (Furnace cooling)	Hardening or Quenching (Air/Oil/Water)
Softer, Less strong	Harder and stronger
More ductile	More brittle
Less internal stress	More internal stress
Less distortion, Cracking	More distortion, Cracking

Selective Hardening

- Carbon steels that have minimum carbon content of 0.4%, or alloy steels with a lower carbon content (hardenable stainless steels with only 0.1% Carbon),
- Selectively hardened in specific regions by applying heat and quench only to those regions.
- These techniques are best suited for medium carbon steels with a carbon content ranging from 0.4 to 0.6%.

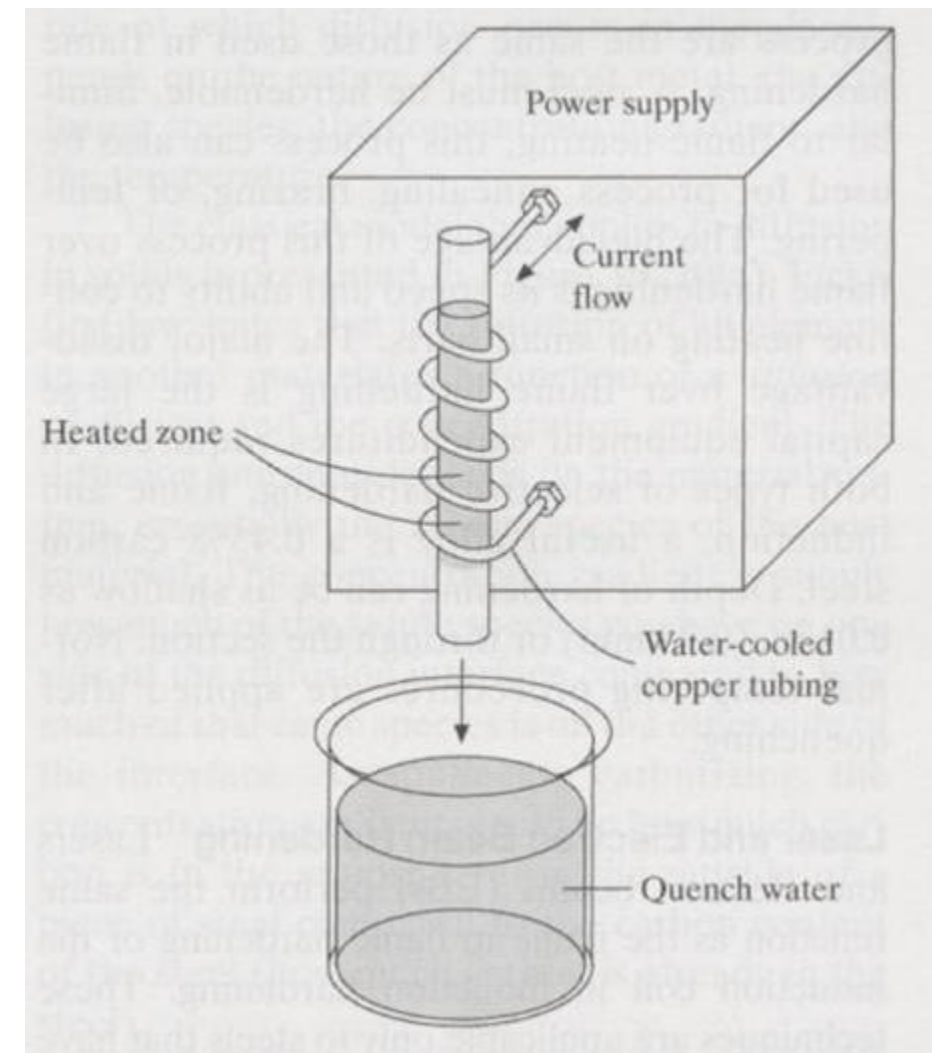
Flame Hardening

- A high intensity oxy-acetylene flame is applied to the selective region.
- Tempering can be done to eliminate brittleness.
- The depth of hardening can be increased by increasing the heating time.
- As much as 6.3 mm (0.25 in) of depth can be achieved.
- In addition, large parts, which will not normally fit in a furnace, can be heat-treated.



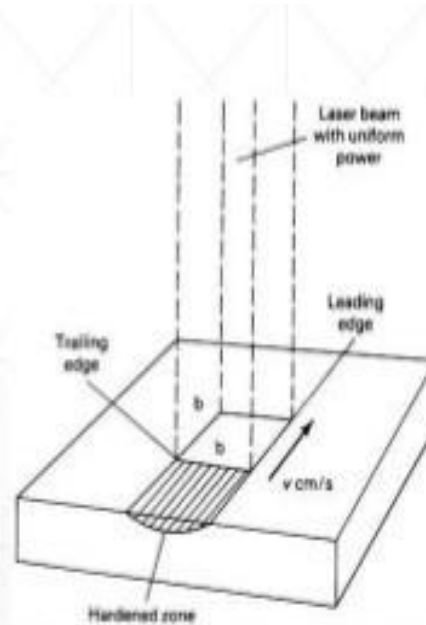
Induction Hardening

- In Induction hardening, the steel part is placed inside a electrical coil which has alternating current through it.
 - This energizes the steel part and heats it up. Depending on the frequency and amperage, the rate of heating as well as the depth of heating can be controlled.
 - The details of heat treatment are similar to flame hardening.
 - Hardness to depth of 0.8mm in 1~5 sec
-
- ✓ Typical applications of induction hardening include gears, shafts, axles, cam lobes, stampings, and spindles, mostly symmetrical parts. Induction hardening is used to strengthen a specific area of a part. Single piece, surface hardening of selective areas.
 - ✓ Carbon steels, Alloy steels, Stainless steels(martensitic), Powder metal, Cast iron, Gray iron, Ductile iron



Laser Beam Hardening

- Laser beam hardening is another variation of flame hardening.
- A phosphate coating is applied over the steel to facilitate absorption of the laser energy.
- By varying the power of the laser, the depth of heat absorption can be controlled.
- The parts are then quenched and tempered.
- This process is very precise and can be run at high speeds, produces very little distortion.
- Case depth 0.75mm.



Electron Beam Hardening

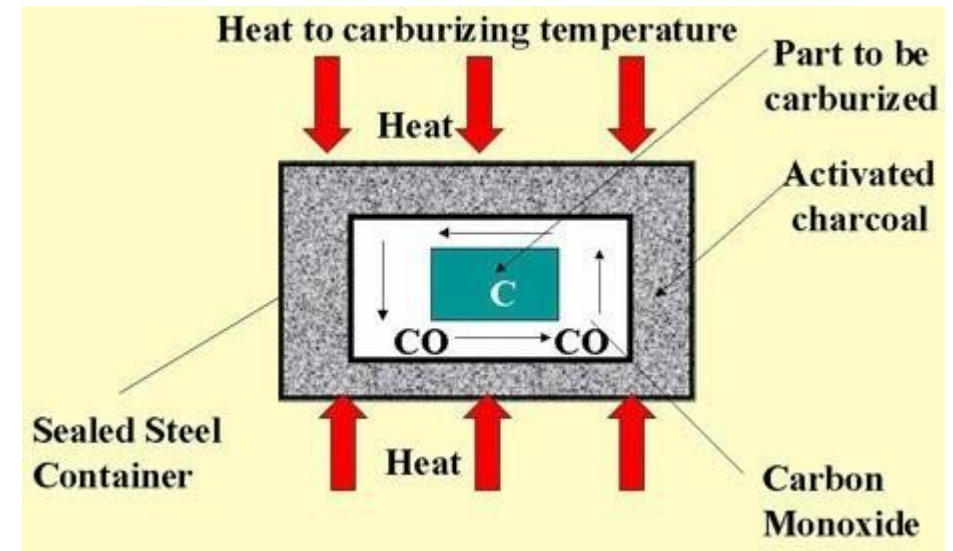
- **Electron beam hardening is similar to laser beam hardening.**
- **The heat source is a beam of high-energy electrons.**
- **The beam is manipulated using electromagnetic coils.**
- **The process can be highly automated, but needs to be performed under vacuum conditions since the electron beams dissipate easily in air.**
- **As in laser beam hardening, the surface can be hardened very precisely both in depth and in location.**
- **Case depth is 0.5~1.5mm.**
- **Automotive transmission clutch.**

Carburizing

- Carburizing also referred to as case hardening, is a heat treatment process that produces a surface which is resistant to wear, while maintaining toughness and strength of the core.
- This treatment is applied to low carbon steel parts after machining, as well as high alloy steel bearings, gears, and other components
- This is done by exposing the part to a carbon rich atmosphere at an elevated temperature and allows diffusion to transfer the carbon atoms into steel.

Pack Carburizing

- Parts are packed in a high carbon medium such as carbon powder or cast iron shavings and heated in a furnace for 12 to 72 hours at 900°C .
- CO gas is produced which is a strong reducing agent.
- The reduction reaction occurs on the surface of the steel releasing carbon, which is then diffused into the surface due to the high temperature.
- The carbon on the surface is 0.7% to 1.2% depending on process conditions.
- The hardness achieved is 60 - 65 RC (Rockwell scale).
- The depth of the case ranges from about 0.1 mm (0.004 in) upto 1.5 mm (0.060 in).
- Difficult to control and inefficient heating.



Gas Carburizing

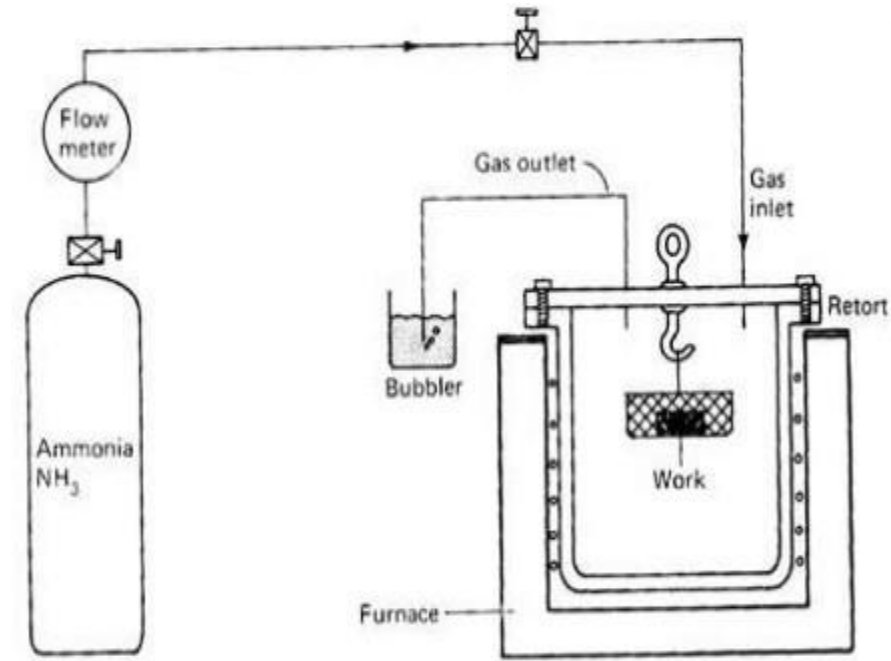
- Gas carburizing is conceptually the same as pack carburizing, except that carbon monoxide (CO) gas is supplied to a heated furnace and the reduction reaction of deposition of carbon takes place on the surface of the part.
- This processes overcomes most of the problems of pack carburizing.
- The temperature diffusion is as good as it can be with a furnace.
- A variation of gas carburizing is when alcohol is dripped into the furnace and it volatilizes readily to provide the reducing reaction for the deposition of the carbon.
- The carbon diffuses into the metal surface usually to a depth between 1 and 3mm

Liquid Carburizing

- The steel parts are immersed in a molten carbon rich bath.
- In the past, such baths have cyanide (CN) as the main component.
- However, safety concerns have led to non-toxic baths that achieve the same result.

Nitriding

- The Nitrogen forms Nitrides with elements such as Aluminum, Chromium, Molybdenum, and Vanadium.
- The parts are heat treated and tempered before nitriding.
- The parts are then cleaned and heated in a furnace in an atmosphere of dissociated Ammonia (containing N and H) for 10 to 40 hours at 500-625 °C.
- Nitrogen diffuses to a depth of upto 0.65 mm (0.025 in).
- The case is very hard and distortion is low.
- No further heat treatment is required; in fact, further heat treatment can crack the hard case.
- Since the case is thin, surface grinding is not recommended.

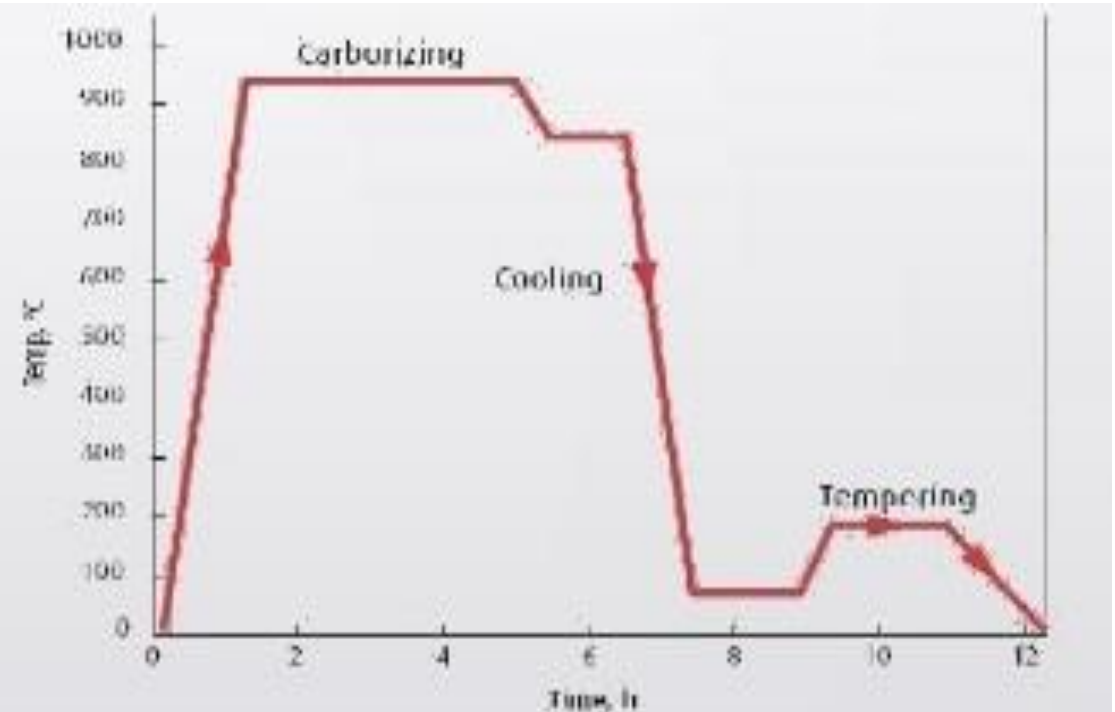
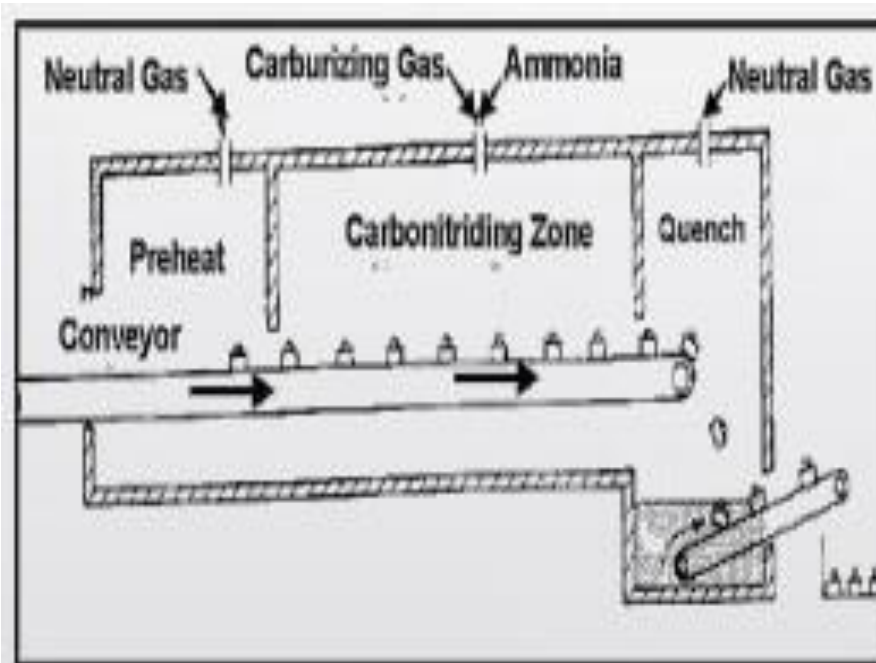


- Typical applications include gears, crankshafts, camshafts, cam followers, valve parts, extruder screws, die- casting tools, forging dies, extrusion dies, firearm components, injectors and plastic-mold tools

Carbonitriding

- Carbonitriding process is most suitable for low carbon and low carbon alloy steels.
- In this process, both Carbon and Nitrogen are diffused into the surface.
- The parts are heated in an atmosphere of hydrocarbon (such as methane or propane) mixed with Ammonia (NH_3)
- The process is a mix of Carburizing and Nitriding.
- Carburizing involves high temperatures (around 900° C) and Nitriding involves much lower temperatures (around 600 ° C).
- Carbonitriding is done at temperatures of 760 - 870 ° C, which is higher than the transformation temperatures of steel that is the region of the FCC Austenite.

- Quenched in a neutral gas (Oxygen free) atmosphere. This quench is less drastic than water or oil, thus less distortion. This process is not suitable for high precision parts due to the distortions that are inherent.
- The hardness achieved is similar to carburizing (60 - 65 RC) but not as high as Nitriding (70 RC). The case depth is from 0.1 to 0.75 mm (0.004 to 0.030 in).



Comparison

